

Week 4 · Heading Control

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	Control Engineering (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	Control System Design (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	Advanced Control Engineering (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	Sensor Signal Processing and Fusion (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	Capstone Design (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

🔧 Getting the course files, and keeping them current (Windows)

The notes, models and scripts are kept in one Git repository that is updated through the semester. Clone it once; before every class, pull. A pull downloads only what has changed since the last one.

When	Where to run it	Command
once	PowerShell — installs Git for Windows	<code>winget install --id Git.Git -e</code>
once	the folder that will hold the course, e.g. <code>Documents</code>	<code>git clone https://github.com/wkyouncnu/Sensor-Signal-Processing-and-Fusion.git</code>
before every class	inside the cloned folder <code>Sensor-Signal-Processing-and-Fusion</code>	<code>git pull</code>

- `git pull` prints `Already up to date.` when nothing has changed, and otherwise lists the files it updated.
- The repository is private. When Git asks, sign in with the GitHub account the instructor has given access.
- Experiment on copies, not on the cloned files: copy a week's `WXX_simulink` folder elsewhere first, and a pull can never collide with local edits. If it already has, `git stash`, then `git pull`, then `git stash pop` sets the edits aside, updates, and puts them back.
- The MSS toolbox is not part of the repository. The weeks that simulate the Otter need it at `Tools\MSS` inside the cloned folder.

- **Course:** USV Guidance, Navigation and Control (Graduate)
- **Department:** Autonomous Vehicle System Engineering, Chungnam National University
- **This week:** ① the Week 2 controller around the Otter's heading, tuned from the Scope with no model of the vessel ② why P alone reaches the command here and D damps again — the opposite of Week 3 ③ the crab angle: heading is not course ④ the moment limit, the back-calculation gain, and the wrap at $\pm 180^\circ$

★ Prerequisites from the previous week

- From Weeks 2 and 3: the five time-domain metrics, what P, I and D do, the filtered derivative, back-calculation anti-windup and the tuning order. Nothing new is added to the controller this week.
- From Week 3 §3-3: the same derivative term that damps a position loop made the speed loop worse. A term's effect is read from the axis, not assumed.
- From Week 1: the two propellers sit $y_{\text{pont}} = 0.395$ m either side of the centreline, so a difference in their thrust turns the vessel.

Learning Outcomes

Upon completion of this week, the learner is able to:

1. Characterise the yaw axis from the outside with one step of yaw moment, and explain from the result why P alone leaves no heading error.
2. Predict, from Weeks 2 and 3 and one fact about the axis, how P, I and D each change the heading response, and confirm it on the Scope.
3. Distinguish heading from course, and compute the crab angle from the body velocities.
4. Remove the heading error left by a weak propeller with the integral, and choose the back-calculation gain from a measured trade-off.
5. Explain why a heading error must be wrapped into $(-180^\circ, 180^\circ]$, and measure the cost of not wrapping it.

Prerequisites and Setup

Item	Requirement
MATLAB	R2024b, with Simulink
MSS	<code>Tools/MSS</code> , found by <code>W04_0_setup</code> through <code>mss_path</code>
Course folder	<code>lectures/W04_simulink</code>
Models	one per example — <code>W04_C_open_loop</code> , <code>W04_D_P</code> , <code>W04_E_PD</code> , <code>W04_F_PID</code> , <code>W04_G_wrap</code> , <code>W04_H_tuning</code> — all generated by <code>W04_1_build_heading</code> , never edited by hand
Time	lecture 40 min, laboratory 1 h 20 min

Part 1 · Theory

4-1. The same controller, a new output

This section answers: what changes when the controller of Weeks 2 and 3 steers the vessel instead of speeding it up?

The controller does not change. The error, the three terms, the derivative filter, the limit and the back-calculation path are the blocks of Week 3, in the same places. Three things around them are new.

New block	What it does	Why
deg to rad	converts the command from degrees to radians	people command headings in degrees; the vessel model works in radians
ssa	wraps the error into $(-\pi, \pi]$: $\text{ssa}(e) = \text{atan2}(\sin e, \cos e)$	-170° is 20° past 170° , not 340° back; §4-6
allocation	a MATLAB Function block, commented line by line: turns the demanded yaw moment N into two propeller thrusts, then two shaft speeds	the Otter has no rudder; it turns by pushing one side harder

The yaw moment comes from a thrust difference. Both propellers push forward to give the surge force X_{ff} ; a yaw moment is added by pushing the port propeller harder and the starboard one less. Each propeller sits y_{pont} from the centreline, so a thrust difference $T_1 - T_2$ gives a moment $y_{\text{pont}}(T_1 - T_2)$, and the two thrusts are

$$T_1 = \frac{X_{ff}}{2} + \frac{N}{2y_{\text{pont}}} \text{ (port)}, \quad T_2 = \frac{X_{ff}}{2} - \frac{N}{2y_{\text{pont}}} \text{ (starboard)}$$

Symbol	Quantity	Value · source
X_{ff}	surge force from both propellers together	60 N, <code>w04_0_setup.m</code>
N	yaw moment demanded by the controller	N·m
y_{pont}	propeller arm, half the distance between the pontoons	0.395 m, <code>otter_config</code>
N_{max}	largest yaw moment available while X_{ff} is delivered	70.85 N·m: the port propeller reaches its 119.68 N maximum at $N = 2y_{\text{pont}}(119.68 - 30)$

- Each thrust is then turned into a shaft speed exactly as in Week 3, $n = \text{sign}(T)\sqrt{|T|/k}$.
- The `moment limit` block holds N inside ± 70.85 N·m. It plays the role the thrust limit played in Week 3, and it causes the windup of §4-5.

4-2. What the vessel does, measured from outside

This section answers: what can one step of yaw moment reveal about the heading?

Model `w04_C_open_loop` has no controller. The vessel runs ahead on $X_{ff} = 60$ N, and at $t = 5$ s a constant yaw moment N is added. Measured in section C:

N [N·m]	final turn rate [deg/s]	turn rate per N·m	time to 63 % of the rate [s]	heading at 40 s [deg]
5	3.05	0.610	0.36	108.2
10	5.06	0.506	0.30	178.3
20	8.14	0.407	0.24	286.0

Three facts follow:

- **The heading never stops.** A constant moment gives a constant turn rate, and the heading grows for as long as the moment is applied. The heading is the running sum — the integral — of the turn rate. In the language of Week 2 §2-4, the plant is **type 1** by itself: a controller needs no integrator of its own to end at the command.
- **The turn rate answers fast.** It reaches 63 % of its final value within **0.36 s** — the vessel starts to turn almost at once, and the slow part is the heading adding up.
- **The turn rate is not proportional to the moment.** Doubling the moment from 10 to 20 N·m raises the rate by only 61 %. The yaw damping grows with the turn rate (Week 1: $N_r(1 + 10|r|)r$), so a quick turn meets more resistance. Unlike the speed of Week 3, this axis is visibly nonlinear — another reason to tune by measurement.

4-3. P, I and D on the heading

This section answers: what does each term do on this axis, and how does it compare with Weeks 2 and 3?

The heading behaves like the position of Week 2 — it can overshoot — but without the spring: nothing pulls the vessel back to a particular heading.

Term	Week 2, position	Week 3, speed	This week, heading	Measured
P	faster, rings, leaves an error	faster, never rings, leaves an error	faster, rings , and leaves no error	section D: overshoot 5.8 → 15.9 % as K_p rises 100 → 1000 ; final heading 10.000°
D	damps	makes it worse	damps again	section E: overshoot 12.18 → 0.39 % , settling 3.26 → 1.72 s as K_d rises 0 → 100 at $K_p = 300$
I	removes the error	removes the error	removes an error only when something pushes back : a weak propeller	section F: 0.80° left by PD with the port propeller at 70 %; removed by $K_i = 20$

- **Why no error with P alone.** At the command the error is zero, so $N = 0$ — and zero moment is exactly what holding a heading takes, because nothing on this axis resists a heading. Compare Week 3: holding a speed took a steady 116 N against the drag.
- **Why D damps.** The derivative of a heading error is a turn rate; resisting a turn rate is damping, as resisting the velocity of the mass was in Week 2.
- **When I is needed.** A propeller that delivers only 70 % of its thrust — fouled, damaged or badly calibrated — makes a steady yaw moment that the controller must oppose for ever. A PD controller opposes it only from error, and settles **0.80°** off. The integral finds the missing moment by itself: **4.15 N·m** at $K_i = 20$, the **4.19 N·m** that PD held with its error.

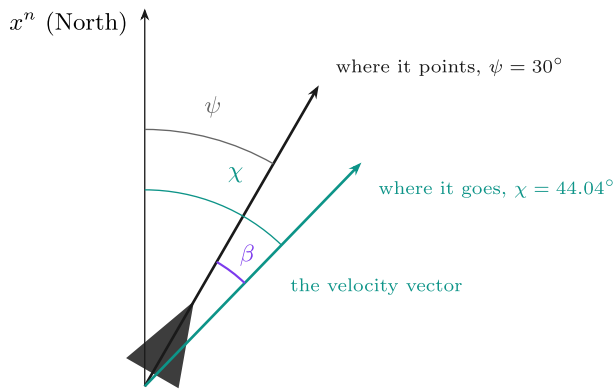
4-4. Heading is not course — the crab angle

This section answers: does a vessel go where it points?

- A heading controller regulates where the vessel **points**. It does not regulate where the vessel **goes**, and those are two different directions.

Heading, course and the crab angle

Where the vessel **points** and where it **goes** are two different directions. A heading controller regulates the first one only.



three angles, one identity

$$\chi = \psi + \beta$$

ψ heading — the hull's x axis, from North

χ course over ground — the velocity, from North

β crab angle — the gap between them, $\beta = \text{atan2}(v, u)$

With $u = 2.0$ and $v = 0.5$: $\beta = 14.04^\circ$ and $\chi = 44.04^\circ$.

β is non-zero whenever the sway velocity v is — in a turn, in a current, in a beam wind. **It is not a disturbance to be rejected; it is what the hull is doing.** Steering ψ while travelling along χ is the permanent path error that Week 4 has to remove, and §4-7 measures it as $\Delta \tan \beta_c$.

In the figure	Meaning
orange ray	the heading ψ — the direction x_b points
blue ray	the course χ — the direction the velocity actually goes
green arc	the crab angle β , the gap between them

$$\beta = \text{atan2}(v, u), \quad \chi = \psi + \beta$$

Symbol	Definition	Unit
ψ	heading, from North to x_b	rad
χ	course over ground, from North to the velocity vector	rad
β	crab angle	rad
u, v	surge and sway velocity (Week 1)	m/s

- With the example of Week 1, $u = 2.0$ m/s and $v = 0.5$ m/s at $\psi = 30^\circ$: $\beta = \text{atan2}(0.5, 2.0) = 14.04^\circ$ and $\chi = 44.04^\circ$.
- The crab angle is non-zero whenever $v \neq 0$ — in a turn, in a current, in a beam wind. It is what the hull is doing, not an error to be removed. Week 1 measured $\beta = -20.3^\circ$ in a turning run of a vessel with no sway actuation at all.
- Fossen writes $\beta = \arcsin(v/U)$; for $u > 0$ it is the same number. The form $\text{atan2}(v, u)$ remains correct going astern.
- This week regulates ψ and lets χ fall where it may. **Week 5 cannot:** a path-following law that steers the heading while the vessel travels along the course leaves a cross-track error of the order of β times the look-ahead distance.

4-5. The tuning order, applied to the heading

This section answers: how are the gains chosen for this axis, without a model — and what does the moment limit add?

The procedure of Week 2 §2-12. The requirement: a 10° turn inside 2 % within 3 s with overshoot below 5 %; no heading error with a weak propeller; no windup in a 90° turn.

Step	What is done	What was measured	Decision
0	requirement; is the plant well behaved?	§4-2: stable turn rate, integrating heading, mildly nonlinear	proceed
1	P only; scale of K_p from the units	a 10° error is 0.175 rad ; $K_p = 300$ asks 52 N·m of the 70.85 available	$K_p = 300$
2	read the P response	overshoot 12.2 % , settling 3.26 s : it rings	add D
3	raise K_d while the response improves	$K_d = 25$: 7.61 % , 2.98 s ; 50 : 4.10 % , 2.22 s ; 100 : 0.39 % , 1.72 s ; 150 : 0 % , 2.42 s	$K_d = 100$, the fastest settling
4	add I if something pushes back	a 70 % port propeller leaves 0.80° ; $K_i = 20$ removes it, settling 1.98 s	$K_i = 20$
5	look at the moment: a big turn	a 90° turn saturates the moment; the back-calculation gain K_b decides what the integral does meanwhile	below

The moment limit and K_b . Measured in section H, settling time and overshoot for three turns:

K_b	10° turn	weak propeller, 10°	90° turn
0 (no anti-windup)	12.42 s, 4.35 %	1.98 s, 0 %	37.12 s, 15.57 %
0.05	7.28 s, 3.20 %	2.18 s, 0 %	24.14 s, 6.38 %
0.1	2.64 s, 2.06 %	9.38 s, 0 %	5.80 s, 0 %
1	33.68 s, 0 %	35.00 s or more, 0 %	48.50 s, 0 %

- **Too little K_b : windup.** While the moment is on its limit the integral keeps charging — to **79.7 N·m** in the **90°** turn — and the vessel overshoots by **15.6 %** and takes 37 s to come back. Even the **10°** turn winds up a little: the derivative kick at the step touches the limit for a moment.
- **Too much K_b : the integral is dragged down.** While saturated, back-calculation pulls the integral towards the value that makes the demand equal to the limit. With $K_p e$ alone asking for several hundred N·m, that value is far below zero — **-249.5 N·m** at $K_b = 1$ — and the vessel stalls short of its heading for tens of seconds. This is the I^* of Week 2 §2-11, on the vessel.
- $K_b = 0.1$ is the compromise: the **90°** turn behaves as if there were no integral at all (**5.80 s** against **5.60 s** for PD alone), the **10°** turn meets the requirement, and the weak propeller is still corrected, more slowly.

The result, $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$, is the default in `w04_0_setup.m`. No transfer function of the vessel was written down.

4-6. The wrap

This section answers: why is the heading error passed through `ssa`?

A heading of **-170°** and one of **170°** are **20°** apart, across the **$\pm 180^\circ$** seam. The raw difference **$-170 - 170 = -340^\circ$** tells the controller to turn **340°** the other way. $\text{ssa}(e) = \text{atan2}(\sin e, \cos e)$ returns the equivalent angle in **$(-180^\circ, 180^\circ]$** , here **$+20^\circ$** .

Measured in section G, the vessel at **170°** commanded to **-170°** at $t = 20$ s:

<code>use_ssa</code>	turn executed after 20 s
1	20.0°
0	340.0° , the other way

- Seventeen times further, for the same commanded heading. Week 5's guidance produces commands anywhere in $(-180^\circ, 180^\circ]$, so the seam is crossed routinely.
-

Part 2 · Laboratory

Every example has **its own model**, laid out as in Weeks 2 and 3: the controller on the left, the thrust split and the vessel in the middle, **one Scope** on the right with the heading in degrees on top and the yaw moment below.

Section	Model	What it shows
C	W04_C_open_loop	a step yaw moment, no controller
D	W04_D_P	P only
E	W04_E_PD	P + D
F	W04_F_PID	P + I + D, with a weak port propeller
G	W04_G_wrap	a command across $\pm 180^\circ$, with an ssa switch
H	W04_H_tuning	the tuning order, a big turn, and the back-calculation gain

Each section ends with an **In class** note: purpose, what to point at, a question with its answer, and the sentence to take away.

A. Setting up (10 min)

```
cd lectures/W04_simulink
W04_0_setup
W04_1_build_heading
```

Expected output:

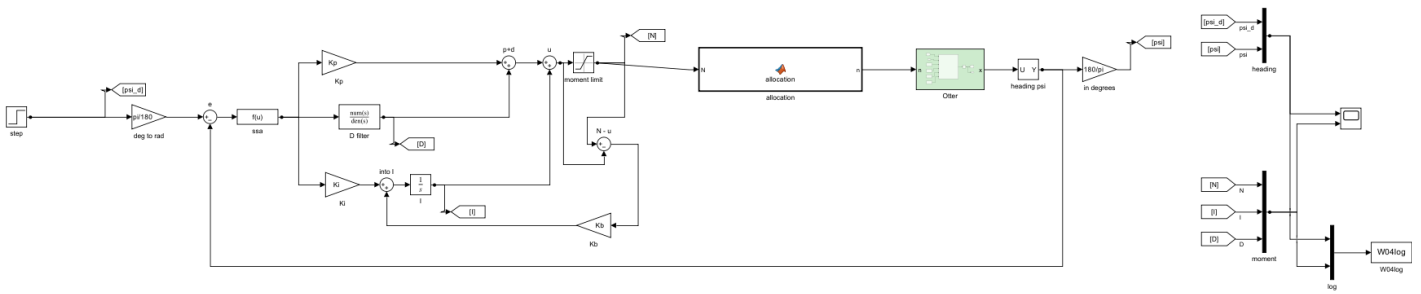
```
W04_0_setup
  thrust   X_ff = 60 N forward, yaw moment |N| <= 70.85 N m
  gains    Kp = 300   Ki = 20   Kd = 100   Nf = 20   Kb = 0.1
  command  psi_d = 0 -> 10 deg at t = 5 s

built W04_C_open_loop.slx (overlapping lines: 0)
built W04_D_P.slx         (overlapping lines: 0)
built W04_E_PD.slx        (overlapping lines: 0)
built W04_F_PID.slx        (overlapping lines: 0)
built W04_G_wrap.slx       (overlapping lines: 0)
built W04_H_tuning.slx     (overlapping lines: 0)
```

- **W04_0_setup.m** is the only file edited by hand. If it stops in **mss_path**, the MSS toolbox is not at **Tools\MSS**.
- The first line of **W04_0_setup** is **clear**. Run it again whenever a Scope does not match these notes.

B. The models (10 min)

```
open_system('W04_H_tuning')
```



WEEK 4, SECTION H - A BIG TURN: THE MOMENT LIMIT AND THE TUNING ORDER

$e = \text{wrap}(\psi_d - \psi)$ in rad, $N = K_p e + K_D \frac{de}{dt} + K_I \int e dt$ (s + M) e
 If N is limited to $\pm N_{\text{max}}$, what the propellers can give while pushing N forward
 anti-windup: $I_{\text{dot}} = K_I e + K_D (N - I)$, $K_D = 0$ switches it off

Change a gain in the Command Window (e.g. $K_p = 300$) and press Run.
 The Scope shows the heading in degrees (psi) and the yaw moment (bottom).

In the figure	Meaning
<code>step</code> → <code>deg to rad</code> → <code>e</code> → <code>ssa</code>	the command in degrees, the error in radians, wrapped
<code>Kp</code> , <code>D filter</code> , <code>Ki</code> → <code>I</code> , <code>p+d</code> , <code>u</code>	the PID of Weeks 2 and 3, block for block
<code>moment limit</code> , <code>N - u</code> , <code>Kb</code>	the moment limit and the back-calculation path of Week 3
<code>allocation</code>	T_1 and T_2 of §4-1, the port propeller's efficiency <code>port_eff</code> (1 unless section F lowers it), and the two shaft speeds; double-click it to read the commented code
<code>Otter</code> → <code>heading psi</code> → <code>in degrees</code>	the vessel, and the heading back in degrees for the Scope
<code>heading</code> , <code>moment</code> → Scope	top: ψ_d and ψ ; bottom: N , I and D

In class

- **Purpose** — show that the controller is still the Week 2 controller; the new parts are the degree conversion, the wrap, and the thrust split that makes a moment out of two propellers.
- **Point to** — open the `allocation` block behind the `moment limit`: one moment in, two thrusts out, one larger and one smaller, each turned into a shaft speed.
- **Ask** — "Why is there a `moment limit` at all, when each propeller has its own limit?" Because the demand must be limited where the controller can see it; otherwise the integral cannot know it was cut, and windup follows (§4-5).
- **Take away** — to steer a rudderless vessel, push one side harder.

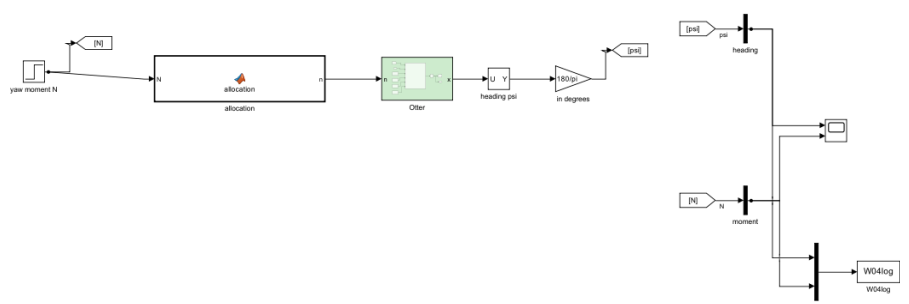
C. The plant seen from outside (10 min)

```
open_system('W04_C_open_loop')
W04_C_plant_from_outside
```

% press Run; then N_open = 20, Run

WEEK 4, SECTION C - THE PLANT, SEEN FROM OUTSIDE (no controller)

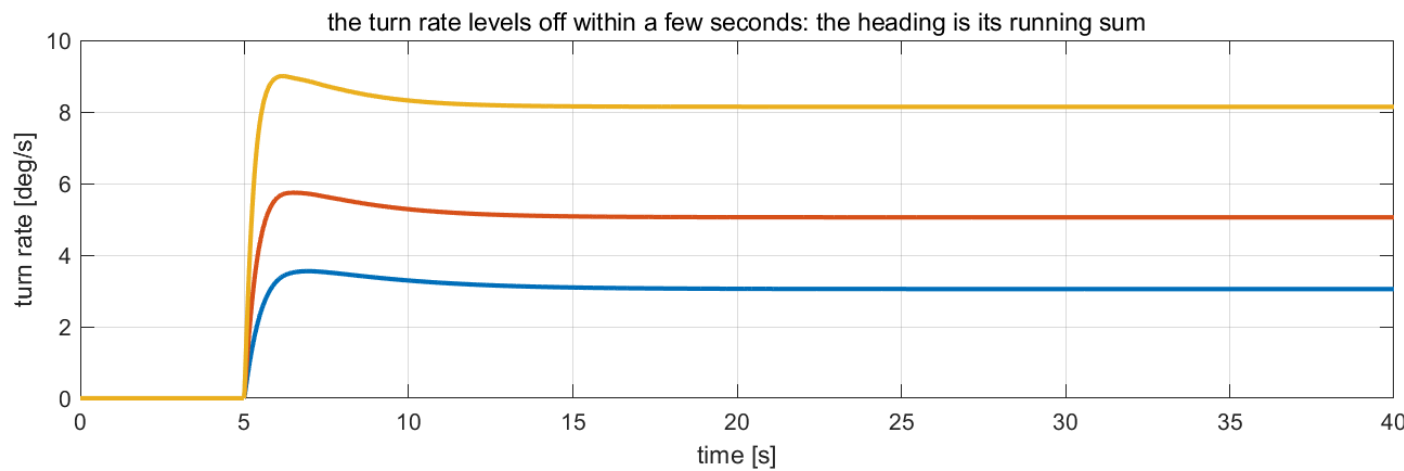
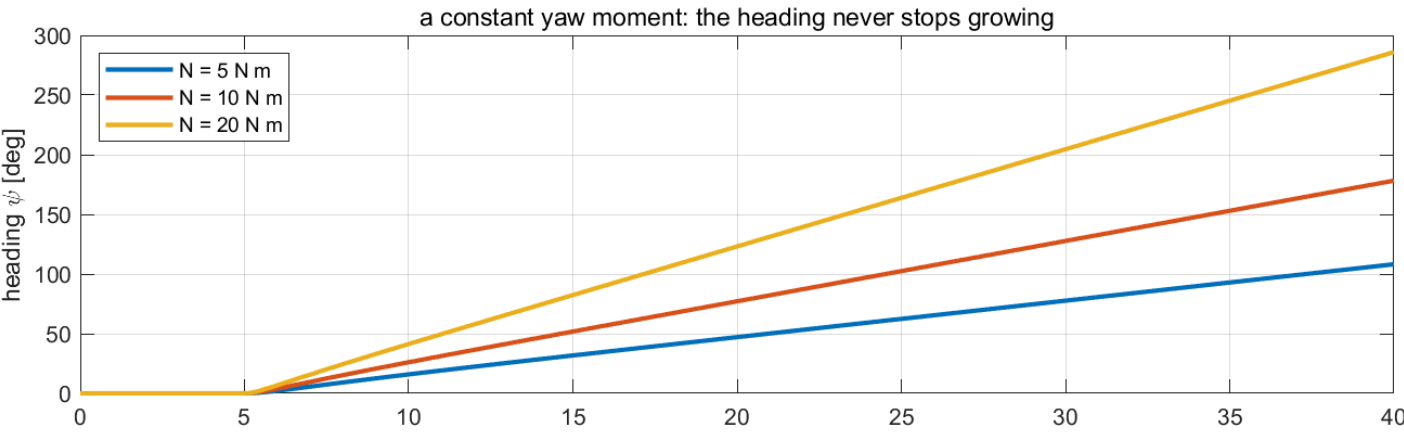
N_{open} is the reaction matrix of yaw moment from $I = I_{\text{stop}}$, with X 5 seconds ahead.
 Change N_{open} (5, 10, 20) and press Run: does the heading ever stop?



In the figure	Meaning
yaw moment N	a step of N_{open} N·m at $t = 5$ s
the thrust split, Otter , heading ψ , in degrees	the plant chain of §4-1
Scope	top: the heading ψ ; bottom: the moment N

Expected output:

W04 C the plant seen from outside (no controller, $X_{\text{ff}} = 60$ N ahead)				
N [N m]	final turn rate [deg/s]	rate per N m	time to 63 % of the rate [s]	heading at 40 s
[deg]				
5	3.05	0.610		0.36
108.2				
10	5.06	0.506		0.30
178.3				
20	8.14	0.407		0.24
286.0				



In the figure	Meaning
top	the heading after a step moment of 5, 10 and 20 N·m
bottom	the turn rate, the slope of the top curves

What the figure says

- The heading grows without end while the turn rate levels off within a few seconds. The heading is the running sum of the turn rate.

In class

- **Purpose** — contrast with Week 3's open loop: there a constant force gave a constant speed; here a constant moment gives a heading that never stops.
- **Point to** — the straight lines in the top panel; the flat rates in the bottom one.
- **Ask** — "What moment does it take to hold a heading?" None: a vessel on a steady heading needs no yaw moment. That is why P alone will leave no error (section D).
- **Ask** — "Why does 20 N·m give less than twice the rate of 10 N·m?" The yaw damping grows with the turn rate.
- **Take away** — the heading is an integral; the plant brings its own.

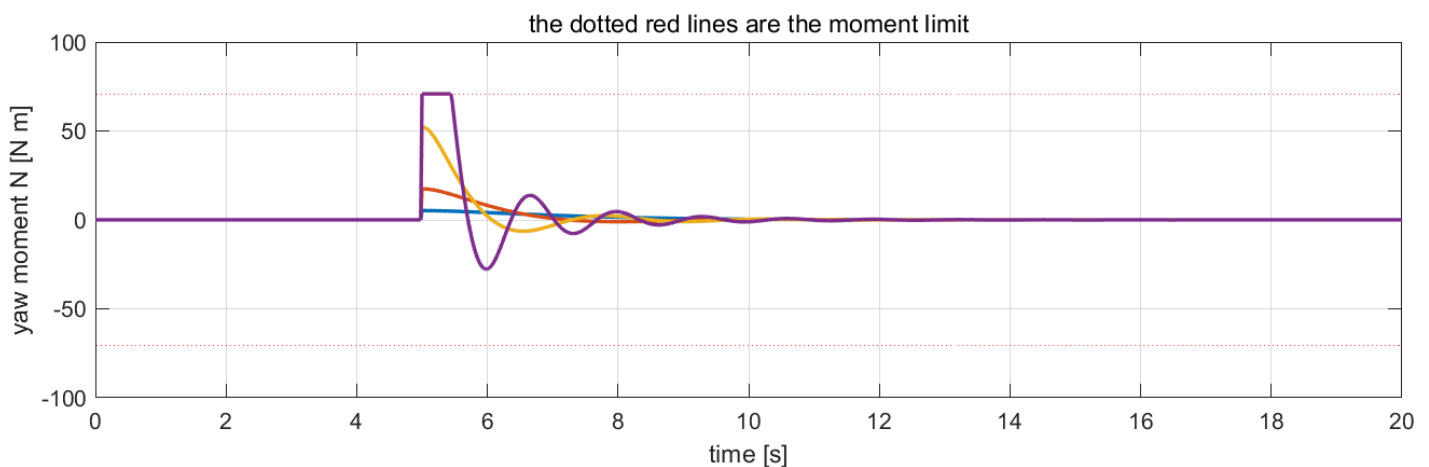
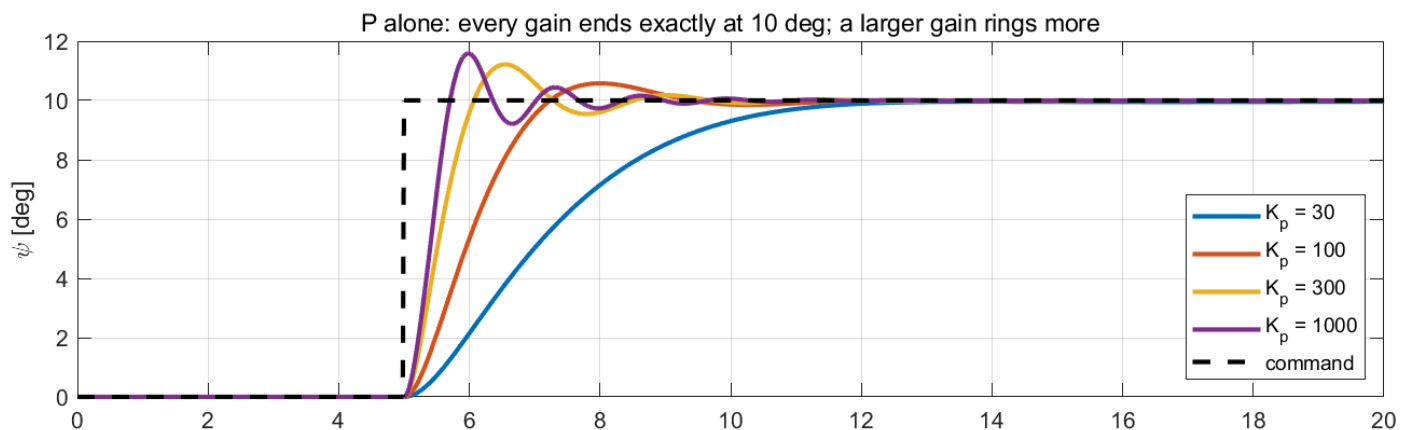
D. P only (10 min)

```
open_system('W04_D_P') % press Run; then Kp = 1000, Run
W04_D_proportional_only
```

Expected output:

W04 D P only (psi_d = 10 deg at t = 5 s)

Kp	final psi [deg]	overshoot %	rise [s]	settle [s]	peak N [N m]
30	9.989	0.0	3.94	6.36	5.2
100	9.997	5.8	1.50	3.96	17.5
300	9.999	12.2	0.74	3.26	52.4
1000	10.000	15.9	0.48	3.12	70.8



In the figure	Meaning
top	the heading for $K_p = 30$ to 1000 and the dashed command
bottom	the yaw moment; the dotted red lines are ± 70.85 N·m

What the figure says

- Every gain ends at 10° ; a larger gain is faster and rings more. At $K_p = 1000$ the moment is on its limit and more gain buys little.

In class

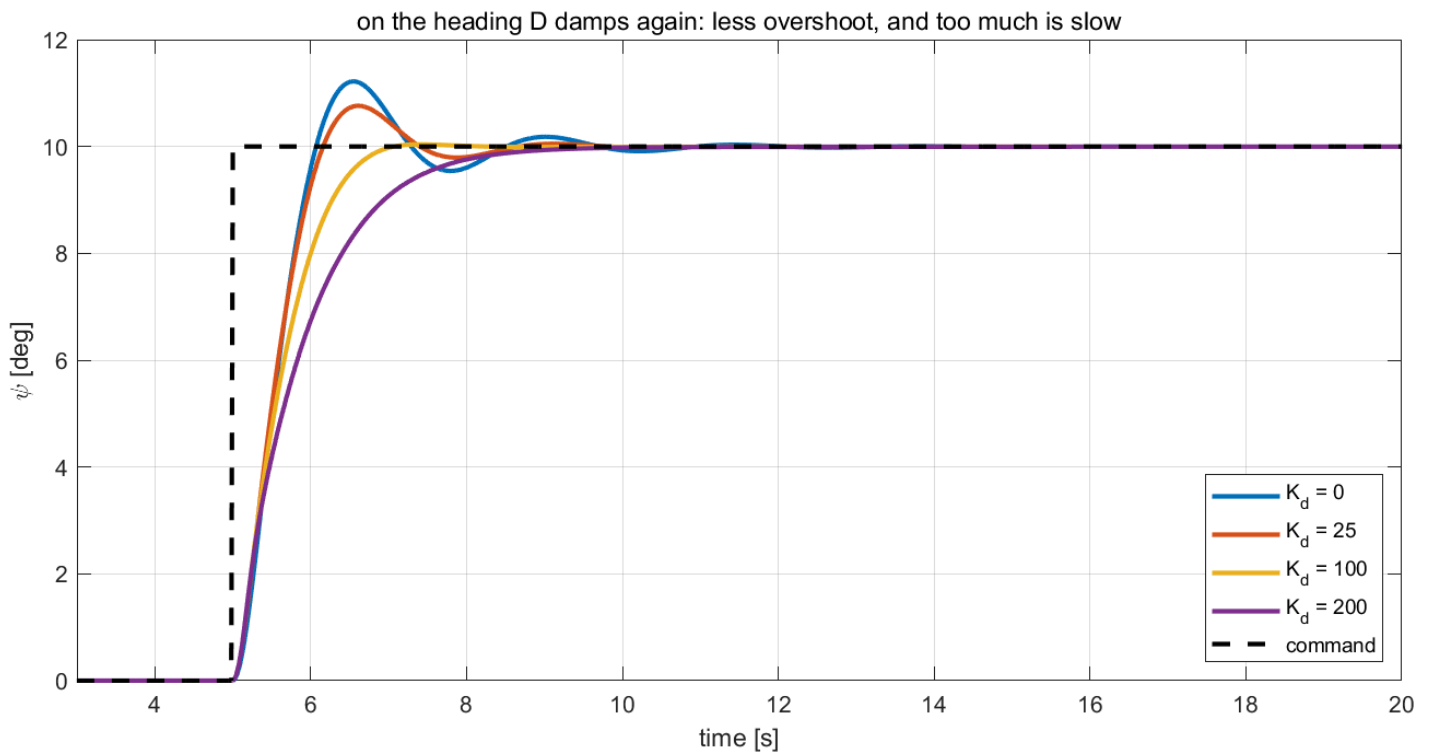
- **Purpose** — the two differences from Week 3 on one plot: no error left, and ringing.
- **Point to** — the final-heading column, all 10° to within 0.011° ; the overshoot column growing with the gain.
- **Ask** — "Week 3 at $K_p = 800$ still missed by 0.133 m/s. Why is this different?" Holding a speed needs force against drag; holding a heading needs no moment.
- **Take away** — on an integrating axis P reaches the command; the question becomes how much it rings.

E. Add D (10 min)

```
open_system('W04_E_PD')           % press Run; then Kd = 0, Run; Kd = 200, Run
W04_E_derivative
```

Expected output:

```
W04 E  P + D  (Kp = 300, psi_d = 10 deg)
  Kd      overshoot %   rise [s]   settle [s]
  0         12.18       0.74       3.26
  10        10.23       0.76       3.20
  25         7.61       0.80       2.98
  50         4.10       0.88       2.22
  100        0.39       1.12       1.72
  150        0.00       1.44       2.42
  200        0.00       1.78       3.14
```



In the figure	Meaning
traces	the heading for $K_d = 0, 25, 100, 200$ at $K_p = 300$

What the figure says

- The derivative removes the ringing; beyond $K_d = 100$ it only slows the turn.

In class

- Purpose** — the derivative on its home ground again, after it failed on the speed loop.
- Point to** — the settling column: it falls to **1.72 s** at $K_d = 100$ and rises again — too much damping is slow (Week 2 §2-3, $\zeta > 1$).
- Ask** — "The same block made Week 3 worse. What changed?" The axis: the derivative of a heading error is a turn rate, and resisting it is damping.
- Take away** — use D where the output can overshoot, and stop raising it where the settling time turns up.

F. A weak port propeller (10 min)

```
open_system('W04_F_PID')           % port_eff = 0.7; Ki = 0; Run; then Ki = 20, Run
W04_F_weak_propeller
```

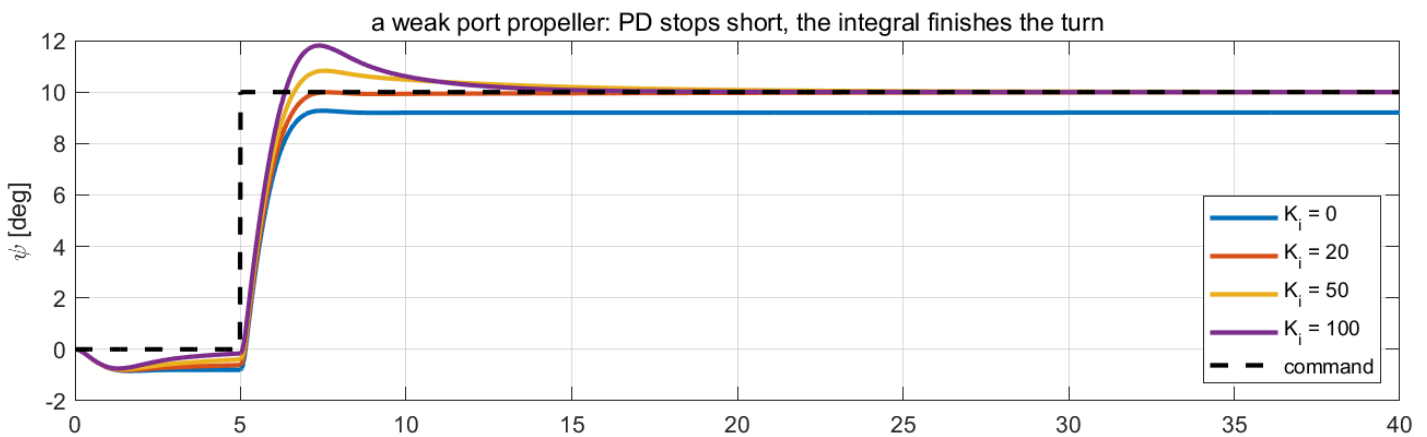
Expected output:

W04 F 1) PD only ($K_p = 300$, $K_d = 100$), port propeller at reduced efficiency

port_eff	final psi [deg]	error left [deg]	steady N [N m]
1	9.999	0.001	0.00
0.7	9.200	0.800	4.19
0.5	8.490	1.510	7.90

2) add the integral, port_eff = 0.7

Ki	final psi [deg]	overshoot %	settle [s]	I at 40 s [N m]
0	9.200	0.00	Inf	0.00
20	9.993	0.00	1.98	4.15
50	10.002	8.27	9.82	4.20
100	10.000	18.07	7.66	4.19



In the figure	Meaning
top	the heading with the port propeller at 70 %, for $K_i = 0$ to 100
bottom	the integral term; every curve ends at the same moment

What the figure says

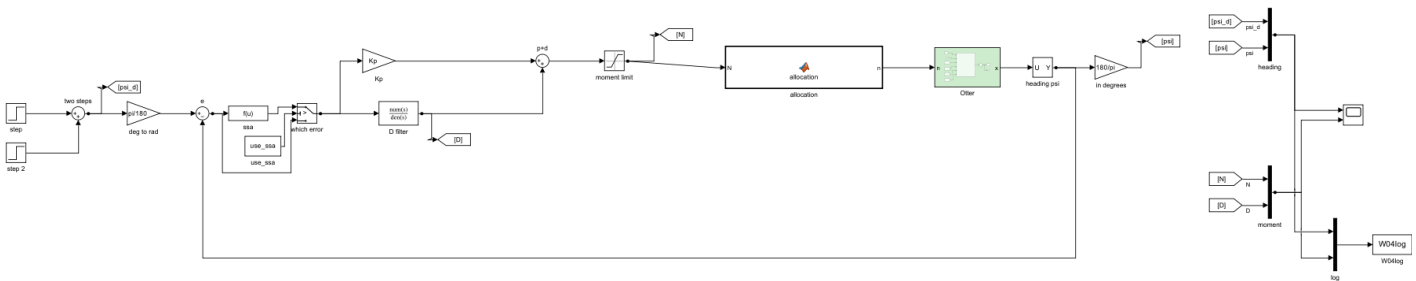
- PD stops 0.80° short, because holding the heading now needs **4.19** N·m that only an error can produce. The integral supplies it and the error goes; too much integral overshoots.

In class

- **Purpose** — show when the integral is needed on an axis that already integrates: when something pushes back.
- **Point to** — the steady N column: **4.19** N·m with PD, the same **4.15–4.20** N·m in the integral once $K_i > 0$.
- **Ask** — "Why does the weak propeller make a yaw moment at all?" Both propellers are asked for 30 N; the port one gives 21 N, so the starboard side pushes harder and the vessel turns to port.
- **Take away** — the integral finds whatever steady effort the plant demands, whether drag (Week 3) or a fault (here).

G. The wrap (10 min)

```
open_system('W04_G_wrap')           % psi_step = 170; psi_step2 = -170; t_step2 = 20; T_final = 6
0; Run; then use_ssa = 0, Run
W04_G_the_wrap
```

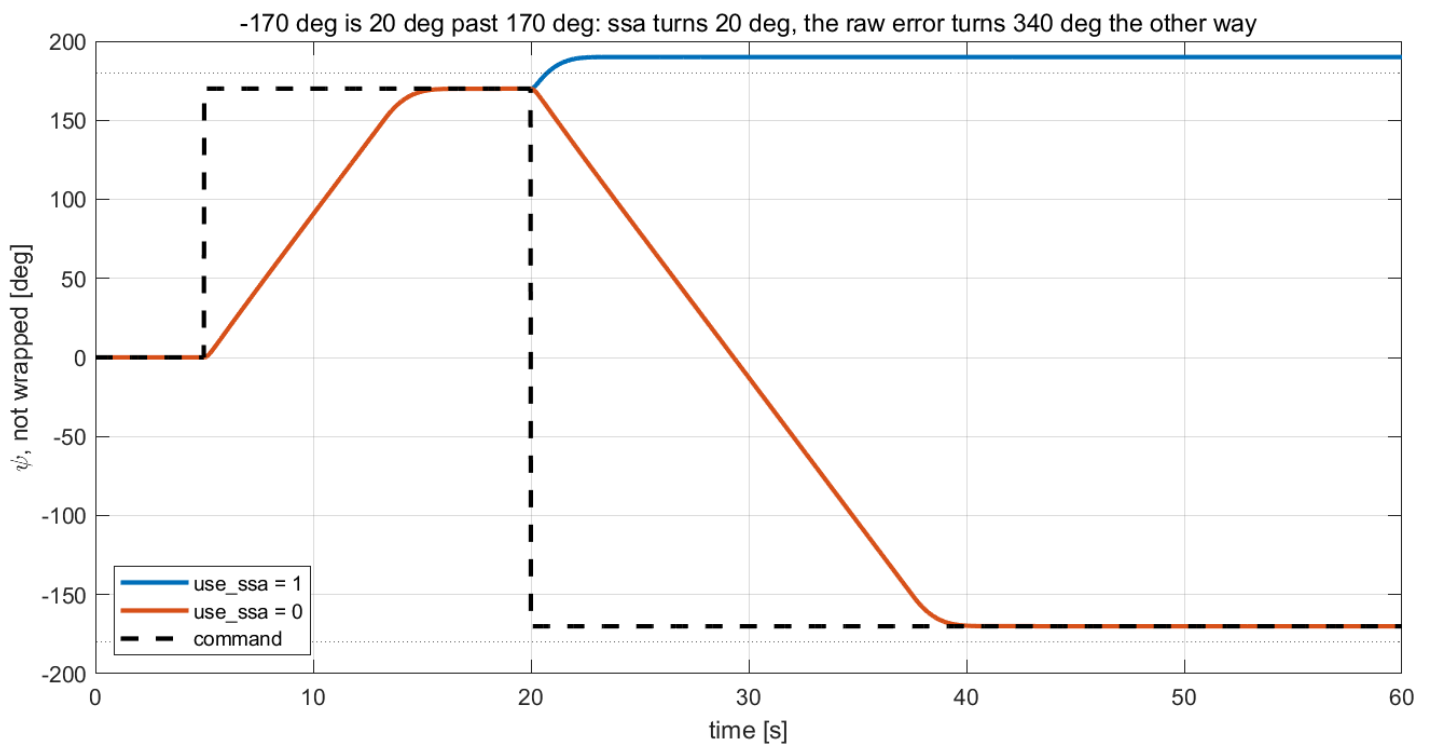


WEEK 4, SECTION G - THE WRAP AT -180 DEG
 e = wrap(psi - psi_d) in rad. N = Kp * e + Ki * (e * T) + Kd * (e * T) * s
 If e is limited to -N_max, what the propellers can give while pointing X, if ahead
 (psi_step = 170; psi_step2 = -170; t_step2 = 20; then use_ssa = 0 and Run again
 Change a gain in the Command Window (e.g. Kp = 300) and press Run.
 The Scope shows the heading in degrees (psi) and the yaw moment (moment).

In the figure	Meaning
ssa, use_ssa, which error	the wrapped error when use_ssa = 1, the raw error when 0
step, step 2, two steps	170° at 5 s, then -170° at t_step2

Expected output:

```
W04 G  170 deg, then -170 deg at t = 20 s
use_ssa  heading at 20 s [deg]  heading at 60 s [deg]  turned after 20 s [deg]
1          170.0                190.0                20.0
0          170.0                -170.0               340.0
```



In the figure	Meaning
traces	the heading, not wrapped, with and without <code>ssa</code> ; the dotted lines are $\pm 180^\circ$

What the figure says

- 190° and -170° are the same heading. With `ssa` the vessel reaches it by a 20° turn; without it, by 340° the other way.

In class

- **Purpose** — one line of code, and the cost of leaving it out.
- **Point to** — the two curves leaving 170° in opposite directions at 20 s.
- **Ask** — "Why does the plotted heading end at 190° and not at -170° ?" The plot shows the heading without wrapping, to make the direction of the turn visible; physically they are the same.
- **Take away** — wrap every heading error before any gain sees it.

H. The tuning order, and a big turn (15 min)

```
open_system('W04_H_tuning')           % psi_step = 90; T_final = 60; Run; then Kb = 0, Run; Kb = 1,
Run
W04_H_tuning_by_hand
```

Expected output:

W04 H the tuning order on the heading

1-2 $K_p = 300$ (10 deg asks 52 N m of the 70.8 available): overshoot 12.2 %, settle 3.26 s: it rings, so D

3 $K_d = 25$: 7.61 %, 2.98 s 50: 4.10 %, 2.22 s 100: 0.39 %, 1.72 s 150: 0.00 %, 2.42 s -> $K_d = 100$ (fastest settling)

4 weak port propeller (0.7): PD leaves 0.80 deg; $K_i = 20$ removes it: 0.007 deg left, settle 1.98 s

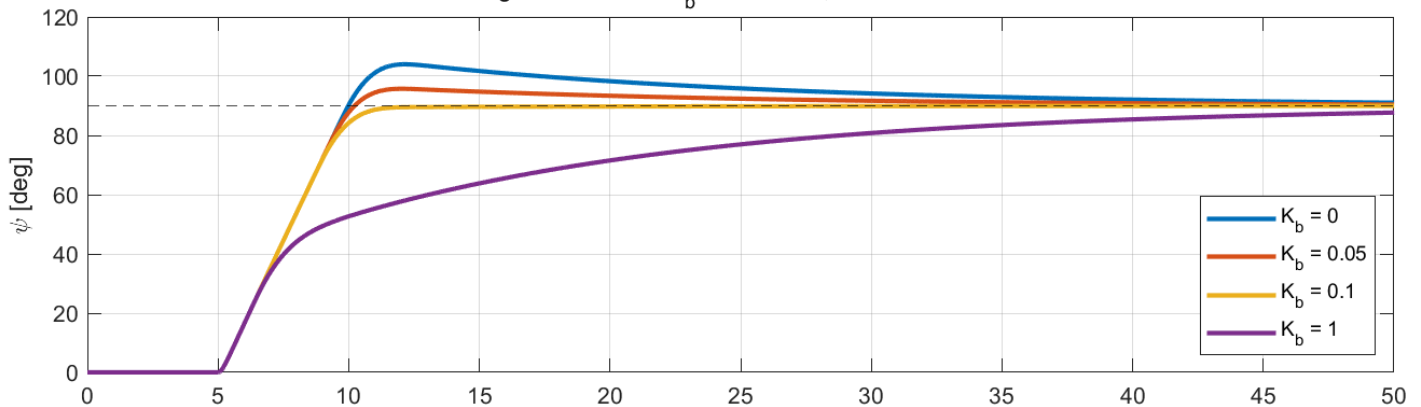
5 the moment limit: K_b against three turns (settle [s], overshoot %)

K_b	10 deg turn		weak propeller		90 deg turn	
0	12.42 s	4.35 %	1.98 s	0.00 %	37.12 s	15.57 %
0.05	7.28 s	3.20 %	2.18 s	0.00 %	24.14 s	6.38 %
0.1	2.64 s	2.06 %	9.38 s	0.00 %	5.80 s	0.00 %
1	33.68 s	0.00 %	35.00 s	0.00 %	48.50 s	0.00 %

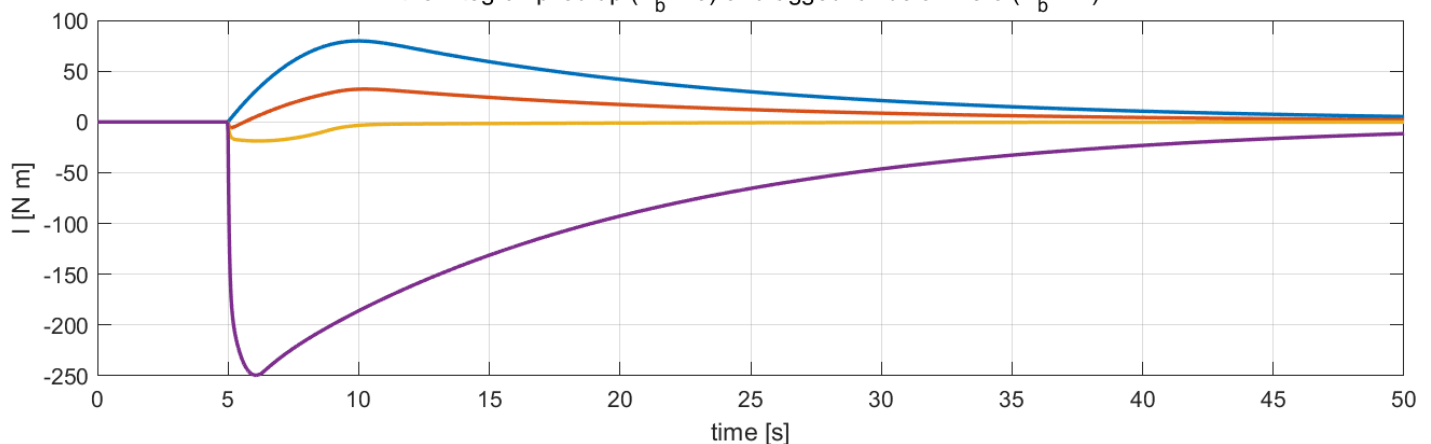
PD alone, 90 deg turn: 5.60 s, 0.00 %

final: $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$

a 90 deg turn: too little K_b overshoots, too much stalls short



the integral: piled up ($K_b = 0$) or dragged far below zero ($K_b = 1$)



In the figure	Meaning
top	the heading in a 90° turn for $K_b = 0, 0.05, 0.1, 1$
bottom	the integral term

What the figure says

- With too little K_b , the integral piles up and the vessel overshoots; with too much it is dragged far below zero and the vessel stalls short. $K_b = 0.1$ turns as cleanly as PD alone.

In class

- **Purpose** — the full procedure, and one decision that cannot be made without looking at the actuator.
 - **Point to** — the blue integral rising to 80 N·m and the purple one diving to **−250** N·m, while the heading curves separate from about 8 s on.
 - **Ask** — "Why does a larger K_b make things worse here, when it helped in Week 3?" In a big turn $K_p e$ alone asks for several hundred N·m; back-calculation drags the integral towards the value that would cancel it, far below zero, and that has to be undone afterwards.
 - **Ask** — "Why is the weak-propeller case slower at $K_b = 0.1$?" The derivative kick at the step touches the limit, back-calculation pulls the integral down, and the integral must then climb to the **4.19** N·m the fault needs.
 - **Take away** — the back-calculation gain is a trade-off, and it is chosen by running the turns that matter.
-

Summary

Week Summary

Step	What was done	How it was verified
1	placed the Week 2 controller around the heading, with <code>ssa</code> and a thrust split	<code>check_overlaps</code> 0 in six models
2	measured the yaw axis from outside	section C: the heading grows without end; turn rate within 0.36 s to 63 %; rate per N-m falls from 0.610 to 0.407
3	P alone	section D: no error at any gain; overshoot 5.8 → 15.9 %
4	D damps	section E: 12.18 → 0.39 %, fastest at $K_d = 100$
5	I against a weak propeller	section F: 0.80 ° removed; integral 4.15 N-m
6	the wrap	section G: 20 ° with <code>ssa</code> , 340 ° without
7	the tuning order and K_b	section H: $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$; 90 ° turn in 5.80 s without overshoot

Progress Check

★ Minimum condition for following Week 5

Theory

- ☐ Able to explain from one open-loop test why P alone leaves no heading error.
- ☐ Able to state what P, I and D do on the heading and compare with Weeks 2 and 3.
- ☐ Able to compute the crab angle and course from u , v and ψ .
- ☐ Able to explain both failure modes of the back-calculation gain.

Laboratory

- ☐ `w04_1_build_heading` ran and every model reported `overlapping lines: 0`.
- ☐ `w04_check(1)`, `(2)` and `(3)` pass on a model built from `w04_P1_start`.

Recorded observations

- ☐ The heading error left by PD with a 70 % port propeller, and the moment the integral found.
- ☐ The **90**°-turn settling time at $K_b = 0, 0.1$ and **1**.

In-class laboratory — build the heading autopilot by hand

The second hour of the Week 4 session is spent building the autopilot in Simulink around a vessel whose allocation is given.

```
cd lectures/w04_simulink/problems
w04_P1_start           % creates w04_P1.slx – hull and allocation only
w04_check(1)          % run this whenever, as often as needed
```

	Problem	Time	The number it must reproduce
1	P only	20 min	no error; overshoot 5.8 % at $K_p = 100$, 12.2 % at 300
2	add the filtered derivative	20 min	overshoot 12.18, 4.10, 0.39 % at $K_d = 0, 50, 100$
3	the wrap	20 min	a 20° turn with ssa , 340° without

- The problem sheet is [w04_simulink/problems/README.md](#), and reference answers are in [w04_simulink/solutions/](#).

Assignment 4

- Due:** before the Week 5 session
- Submit:** the modified [w04_0_setup.m](#), the numbers requested below, and two figures

① Requirements

Halve the surge force, $X_{ff} = 30$ N, in [w04_0_setup.m](#) and [w04_vars.m](#), and tune the heading loop again by the tuning order of §4-5 with the same requirement.

② Verification — mandatory

★ A claim is not a result. Numbers are required, with the tolerance band stated

- The new $N_{\max}$, computed from §4-1 and read from [w04_0_setup](#)'s output.
- Section C with the new X_{ff} : the turn rate per N·m at 10 N·m.
- For each step of the tuning order, the measurement that decided it.
- The K_b table of §4-5 for the final gains.

③ Analysis (5–10 lines)

Explain why the available yaw moment depends on the surge force, and how that moved the gains.

Grading

Criterion	Weight
The model runs and produces the requested output	25%
Verification performed and numbers reported	40%
Correctness of the analysis	25%
Readability of the code	10%

Troubleshooting

Symptom	Cause	Fix
<code>w04_0_setup</code> stops in <code>mss_path</code>	the MSS toolbox is not at <code>Tools\MSS</code>	place MSS there; see the first page
the vessel stalls short of a big turn for tens of seconds	K_b too large	<code>Kb = 0.1</code>
a big turn overshoots and creeps back	$K_b = 0$: windup	<code>Kb = 0.1</code>
the vessel turns the long way round	<code>use_ssa = 0</code>	<code>use_ssa = 1</code>
a Scope does not match these notes	a changed variable is still in the workspace	run <code>w04_0_setup</code> again
<code>allocation</code> reports that <code>x_ff</code> or <code>k_pos</code> is undefined	<code>w04_0_setup</code> was not run, so the block's parameters have no values	run <code>w04_0_setup</code> ; the block reads <code>x_ff</code> , <code>y_pont</code> , <code>k_pos</code> , <code>k_neg</code> and <code>port_eff</code> from the workspace

References

Primary

- Fossen, T. I. *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., Wiley, 2021, §12.2 (PID control of marine craft) and §12.2.6 (anti-windup); the crab angle and course are defined with the kinematics of Ch. 2.
- Åström, K. J. and Hägglund, T. *Advanced PID Control*, ISA, 2006, Ch. 3.
- MSS toolbox, `Tools/MSS/VESSELS/otter.m` and `GNC/ssa.m`.

In this course

- Week 2 §2-4, §2-6 to §2-8, §2-11 and §2-12; Week 3 §3-3 and §3-5.
- The previous, model-based version of this week — the Nomoto model, pole placement and the second-order formulas — is kept in `w04_simulink/_previous_version/` for reference.

Next Week

- Week 5 — Waypoint Following and LOS Guidance**
- A heading autopilot follows a commanded heading; guidance decides what heading to command. Week 5 builds the line-of-sight law that turns a list of waypoints into ψ_d , and meets the crab angle of §4-4 as a cross-track error.